Containerization-and-DevOps

🐳 Java Docker App – Class Task

🛠️ Technologies Used

- Docker
- Ubuntu 22.04 (Base Image)
- OpenJDK 17
- Java

📂 Project Structure

```
java-docker/
├── Dockerfile
├── Hello.java
```

📄 Dockerfile

```
FROM ubuntu:22.04

RUN apt update && apt install -y openjdk-17-jdk

WORKDIR /home/app

COPY Hello.java .

RUN javac Hello.java

CMD ["echo", "Hello from version 2"]
```

⚙️ Steps Performed

1️⃣ Build Docker Image

```
docker build -t java-app:1.0 .
```

2️⃣ Run the Container

```
docker run java-app:1.0
```

Output:

```
Hello from Docker!
```

3️⃣ Run Container in Interactive Mode

```
docker run -it java-app:1.0 bash
```

4️⃣ Login to Docker Hub

```
docker login -u sristee
```

> Used Personal Access Token (PAT) instead of password.

5️⃣ Tag the Image

```
docker tag java-app:1.0 sristee/java-app:1.0
```

6️⃣ Push Image to Docker Hub

```
docker push sristee/java-app:1.0
```

🚀 Docker Hub Repository

Image successfully pushed to:

```
docker.io/sristee/java-app
```

⚠️ Notes

- Docker may show a warning about storing credentials unencrypted.
- Build used cached layers for faster execution.
- CMD was modified to print a custom message (`Hello from version 2`).

✅ Outcome

- Successfully created and built a Docker image
- Ran containers using the image
- Authenticated with Docker Hub using PAT
- Tagged and pushed the image to a remote repository

📸 Screenshots

> all screenshots below:

```
Successfully tagged java-app:1.0
sristee@Sristee:/mnt/c/Users/srist/java-docker$ docker run java-app:1.0
Hello from Docker!
sristee@Sristee:/mnt/c/Users/srist/java-docker$ docker run -it java-app:1.0 bash
root@f6171aa1a470:/home/app#
```

```
sristee@Sristee: /mnt/c/Users
  GNU nano 7.2                                    Dockerfile *
FROM ubuntu:22.04

RUN apt update && apt install -y openjdk-17-jdk

WORKDIR /home/app

COPY Hello.java .

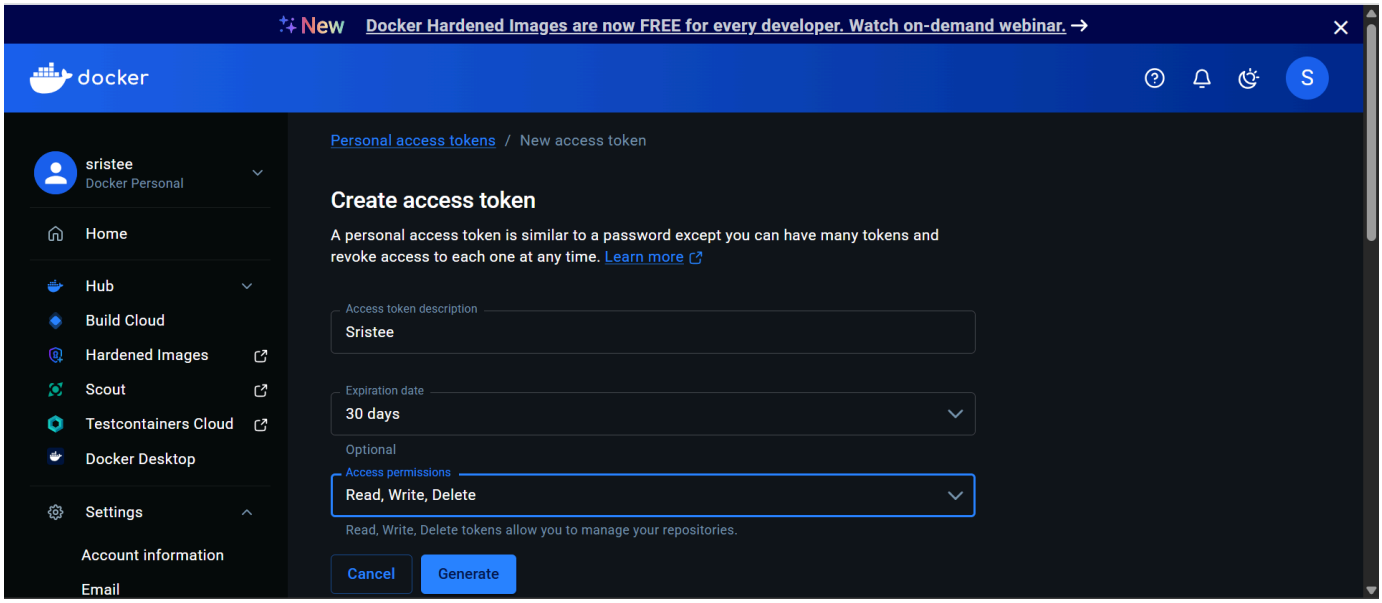
RUN javac Hello.java

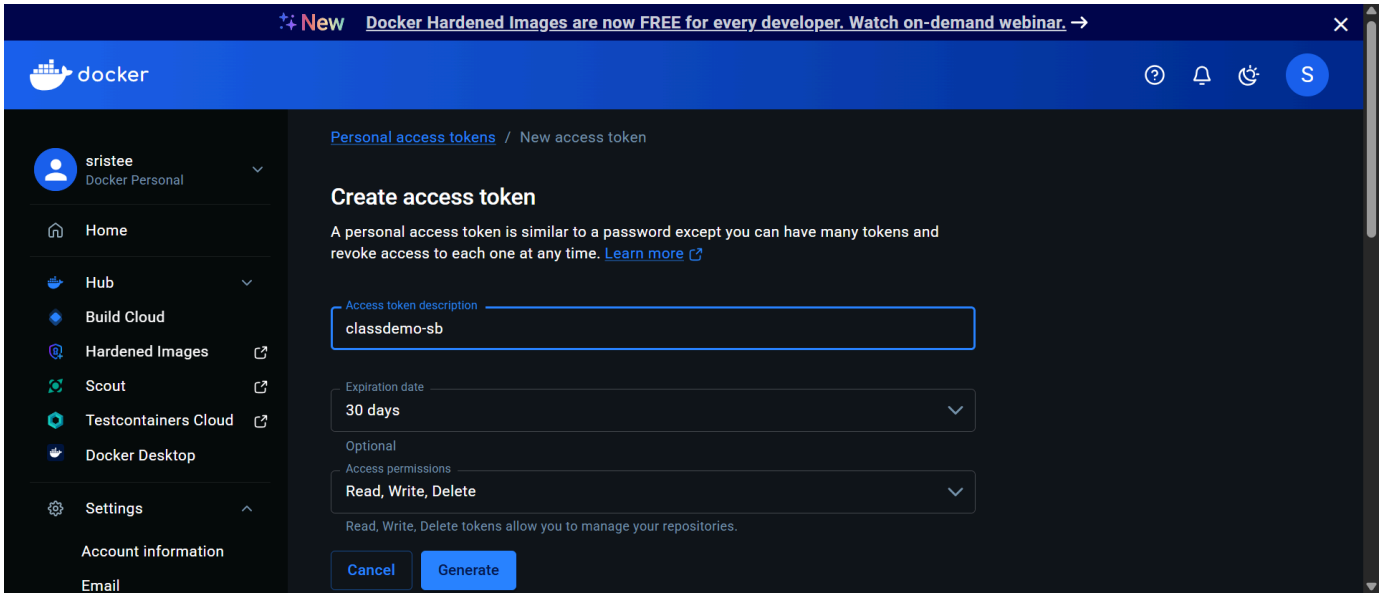
CMD ["echo", "Hello from version 2"]

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo    M-6 Copy
```

```
sristee@Sristee:/mnt/c/Users/srist/java-docker$ nano Dockerfile
sristee@Sristee:/mnt/c/Users/srist/java-docker$ docker run java-app:1.0
Hello from Docker!
sristee@Sristee:/mnt/c/Users/srist/java-docker$ docker build -t java-app:1.0 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/6 : FROM ubuntu:22.04
 ---> 65c77cbc27c2
Step 2/6 : RUN apt update && apt install -y openjdk-17-jdk
 ---> Using cache
 ---> fd59dfa365c9
Step 3/6 : WORKDIR /home/app
 ---> Using cache
 ---> 02f0bdb5ef3a
Step 4/6 : COPY Hello.java .
 ---> Using cache
 ---> e2ece0c94106
Step 5/6 : RUN javac Hello.java
 ---> Using cache
 ---> 0b251f81af10
Step 6/6 : CMD ["echo", "Hello from version 2"]
 ---> Running in c01646c1209c
 ---> Removed intermediate container c01646c1209c
 ---> 11efb5c08422
Successfully built 11efb5c08422
Successfully tagged java-app:1.0
sristee@Sristee:/mnt/c/Users/srist/java-docker$
```


New Docker Hardened Images are now FREE for every developer. Watch on-demand webinar. →
docker
S
sristee
Docker Personal
Home
Hub
Build Cloud
Hardened Images
Scout
Testcontainers Cloud
Docker Desktop
Settings
Account information
Email
Personal access tokens / New access token
Create access token
A personal access token is similar to a password except you can have many tokens and revoke access to each one at any time. Learn more
Access token description
Sristee
Expiration date
30 days
Optional
Access permissions
Read, Write, Delete
Read, Write, Delete tokens allow you to manage your repositories.
Cancel
Generate


New Docker Hardened Images are now FREE for every developer. Watch on-demand webinar. →
docker
S
sristee
Docker Personal
Home
Hub
Build Cloud
Hardened Images
Scout
Testcontainers Cloud
Docker Desktop
Settings
Account information
Email
Personal access tokens / New access token
Create access token
A personal access token is similar to a password except you can have many tokens and revoke access to each one at any time. Learn more
Access token description
classdemo-sb
Expiration date
30 days
Optional
Access permissions
Read, Write, Delete
Read, Write, Delete tokens allow you to manage your repositories.
Cancel
Generate

```
sristee@Sristee:/mnt/c/Users/srist/java-docker$ docker login -u sristee

i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:
dckr_pat_C3wHnqgGn7oaT8mAhDv1G0Tvd2g

WARNING! Your credentials are stored unencrypted in '/home/sristee/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
sristee@Sristee:/mnt/c/Users/srist/java-docker$ dckr_pat_C3wHnqgGn7oaT8mAhDv1G0Tvd2g
dckr_pat_C3wHnqgGn7oaT8mAhDv1G0Tvd2g: command not found
sristee@Sristee:/mnt/c/Users/srist/java-docker$
```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Login Succeeded
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dckr_pat_C3wHnqgGn7oaT8mAhDv1G0Tvd2g: command not found
```


```
sristee@Sristee:/mnt/c/Users/srist/java-docker$ docker tag java-app:1.0 sristee/java-app:1.0
sristee@Sristee:/mnt/c/Users/srist/java-docker$ docker push sristee/java-app:1.0
The push refers to repository [docker.io/sristee/java-app]
3c1d34925256: Pushed
2b4e502fe2e2: Pushed
7e502cc71122: Pushed
3e281eef7df1: Pushing [===>                                                ]  43.53MB/695.1MB
fbb9bbbaf4d2: Mounted from library/ubuntu
```